

consumed by a service instance which leads to efficient utilisation of resources.

- Greater consistency across different environments is achieved.

Let us look at how to build a simple microservice. This will be a Node.js application that will run within a Docker container. Our app will be a simple 'Hello world' app and the source code for it can be downloaded from <https://github.com/chetanarale/DockerMicroservices>.

 **Note:** All the examples in this tutorial are demonstrated on Ubuntu.

The first step is to install Docker. To do so on Ubuntu, follow the instructions given at <https://docs.docker.com/engine/installation/linux/docker-ce/ubuntu>. Once Docker is installed, we will pull an Ubuntu image from the Docker hub using the command given below:

```
[bash$] docker pull library/ubuntu

Using default tag: latest
latest: Pulling from library/ubuntu
6b683210e931: Pull complete
2710139890b6: Pull complete
73f656e3a66: Pull complete
0153a4475206: Pull complete
ec00daa85fe6: Pull complete
2ebbd207dc43f: Pull complete
Digest: sha256:8e6b67faf44a036a18b9a3987e826fe3accbd1742e219
145a461a3210a9d8142
Status: Downloaded newer image for ubuntu:latest
```

This will be our base image, and we will install everything required to run our application on this image. Next, we will install Node.js on our image using Dockerfile, which contains commands that will be run when building a container. Here is how our Dockerfile looks:

```
Dockerfile

FROM ubuntu:latest
ENV DEBIAN_FRONTEND noninteractive
ADD . /home/microservice/
RUN /home/microservice/installScript.sh
```

This Dockerfile tells Docker to use the newly pulled Ubuntu image — if the image was not already pulled, then it first pulls the Ubuntu image from the registry. The `ADD` command gives the instruction to copy the contents of the current directory to the directory named 'microservice' under `/home`. The `RUN` command executes the script named `installScript.sh`. This file contains the command required to install Node.js.

```
installScript.sh
```

```
apt-get update
apt-get install nodejs
apt-get install npm
```

The next step is to build the image from the Dockerfile, for which the command is:

```
[bash$] docker build -t "mservice/base:latest"
```

When the build successfully completes, a new Docker image is created with the name `mservice`. To view all the images present on the machine, run the `Docker images` command. This newly created image has Node.js installed in it and can be used as the base image for all Node.js applications from now on. You can also push the image to Docker-hub.

The next step is to build our app from the `mservice/base` image. The Dockerfile is as follows:

```
Dockerfile

FROM mservice/base:latest
ADD . /home/microservice/app
ENTRYPOINT ["/home/microservice/app/boot.sh"]
```

On the previously created base image, we copy the contents of our app to the directory `/home/microservice/app` and then execute the file `boot.sh`, which has one simple command to start our node app. Now that we have built our app within a container, the next step is to run the container. Use the following command to run the app:

```
docker run -d --name=nodes -p 3000:3000
```

The `-d` flag instructs Docker to run the container in the background, `--name` assigns a name to the container, and the `-p` flag specifies the `host:container` port mapping.

We now have our app running within a container, and making a Curl request to `localhost:3000` gets the response 'Hello World!'

The service that we just built looks like any other application running in a container, but the key to building a good microservice is carefully breaking down an application into smaller components. This makes the application easy to maintain. The many advantages of using Docker to build microservices definitely complement this style of development. 

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